

Haptic Glove

Technical Documentation & Assembly Guide

20-channel wireless vibration motor wearable

Project	20-motor haptic feedback glove
Control	ESP32 with WiFi web server
Architecture	3× 74LS595 shift registers + 3× ULN2803A drivers
Power (control)	5V via USB or boost converter
Power (motors)	3.7V from 18650 Li-ion battery
Form factor	Wrist-mounted control board + glove-mounted motors
Assembly options	Perfboard (hand-soldered) or custom PCB

This document contains the complete technical specification for assembling the haptic glove: system architecture, detailed wiring schematics, full bill of materials, every pin-by-pin connection, and step-by-step assembly notes.

1. System Overview

The haptic glove uses an ESP32 microcontroller to drive 20 individual vibration motors arranged across the fingers, thumb, and palm of a glove. Communication is wireless via WiFi: a web GUI on any device on the same network sends commands to the ESP32 to activate motors in any pattern.

1.1 Signal flow

The ESP32 controls 20 outputs using only 3 GPIO pins by cascading three 74LS595 shift register chips. The shift registers send small logic signals; these signals drive ULN2803A Darlington arrays, which act as power amplifiers capable of switching the motor currents. Each motor is powered from a separate 3.7V battery rail, and is switched on by the corresponding ULN2803A output sinking current to ground.

1.2 Power separation (critical)

Two separate power rails are used: one for the control electronics (5V) and one for the motors (3.7V from the Li-ion battery). These rails must NEVER be connected to each other. Only the ground (GND) is shared between them.

Why? Motors generate current spikes and electrical noise when starting and stopping. If the ESP32 shared the same +V rail as the motors, the voltage would sag every time motors fire, causing the ESP32 to brown-out, reset, or lose WiFi.

1.3 Architecture diagram

ESP32 (USB power) → 3× 74LS595 (cascaded) → 3× ULN2803A → 20 motors → 18650 battery (separate rail)

2. Bill of Materials (BOM)

2.1 Required components

#	Component	Qty	Notes
1	ESP32 dev board	1	ESP32-WROOM-32, with WiFi (e.g. DevKit V1)
2	SN74LS595N or 74HC595	3	DIP-16 package, 5V logic
3	ULN2803A	3	DIP-18 package, Darlington array
4	Coin vibration motors	20	3-5V rated, ~80mA each, 8-10mm dia
5	18650 Li-ion cell	1	3.7V, prefer protected cell
6	18650 battery holder	1	With wire leads
7	SPST slide switch	1	Inline on battery (+) wire
8	DIP-16 IC sockets	3	Optional, for 74LS595s
9	DIP-18 IC sockets	3	Optional, for ULN2803As
10	100nF ceramic capacitors	6	Decoupling, near each IC
11	10uF electrolytic capacitors	2	Bulk decoupling, one per rail
12	Perfboard or custom PCB	1	~50x70mm minimum
13	Male header pins (strip)	1	For ESP32 connection
14	Ribbon cable (22 conductors)	30 cm	20 motor lines + GND + spare
15	Solder, flux, hookup wire	—	Standard supplies

Note on chips: 74HC595 is preferred over 74LS595N because it draws less power and runs cooler, but both work identically in this circuit. If using 74HC595, run it at 5V — same wiring.

2.2 Recommended additional components (future-proof)

Component	Purpose
TP4056 charging module	USB recharging for the 18650 battery
MT3608 boost converter (3.7V → 5V)	Run ESP32 from battery for full wireless operation
Status LED + 1kΩ resistor	Visual power-on indicator
Pin headers (extra)	Breakout for adding sensors later

3. Pin Connections

3.1 ESP32 to 74LS595 chain

ESP32 Pin	Function	Connects To
VIN (or 5V)	Power input	CHIP +5V rail
GND	Ground	Shared GND rail
GPIO 23	DATA (SER)	74LS595 #1, pin 14
GPIO 19	CLOCK (SRCK)	All 3× 74LS595, pin 11 (shared bus)
GPIO 21	LATCH (RCK)	All 3× 74LS595, pin 12 (shared bus)

Important: GPIO 23 (data) connects to chip #1 ONLY. Chips #2 and #3 receive data through the daisy chain from the previous chip's pin 9, not from the ESP32.

3.2 74LS595 support pins (apply to all 3 chips)

Pin	Name	Connect To	Notes
16	Vcc	CHIP +5V rail	Power supply
8	GND	Shared GND rail	Ground
10	SRCLR	CHIP +5V rail	HIGH = no reset (always)
13	G	Shared GND rail	LOW = outputs enabled (always)
11	SRCK	Shared clock bus → ESP32 GPIO 19	Shift clock
12	RCK	Shared latch bus → ESP32 GPIO 21	Output latch

3.3 74LS595 daisy chain

From	To	Notes
74LS595 #1 pin 9 (Q_H')	74LS595 #2 pin 14 (SER)	First chain link
74LS595 #2 pin 9 (Q_H')	74LS595 #3 pin 14 (SER)	Second chain link
74LS595 #3 pin 9	(unused)	Leave disconnected

3.4 ULN2803A support pins (apply to all 3 chips)

Pin	Name	Connect To	Notes
9	GND	Shared GND rail	Ground reference
10	COM	MOTOR +3.7V rail	CRITICAL: motor power, not chip power

Most common assembly mistake: ULN2803A pin 10 (COM) must connect to the MOTOR +3.7V rail, NOT the chip +5V rail. This pin powers the internal flyback diodes that protect the chip from motor voltage spikes. Wrong connection causes random failures and possible chip damage.

3.5 74LS595 outputs to ULN2803A inputs

Direct wires, no resistors. Each 74LS595 chip pairs with one ULN2803A.

74LS595 Pin	Label	→ ULN2803A Pin	Label
15	Q_A	1	1B
1	Q_B	2	2B
2	Q_C	3	3B
3	Q_D	4	4B
4	Q_E	5	5B
5	Q_F	6	6B
6	Q_G	7	7B
7	Q_H	8	8B

Apply this pattern for all 3 pairs (595#1↔ULN#1, 595#2↔ULN#2, 595#3↔ULN#3). For the 595#3↔ULN#3 pair, only the first 4 wires are used (pins 15, 1, 2, 3 → 1, 2, 3, 4) since chip #3 only drives 4 motors.

3.6 Complete motor wiring map (all 20 motors)

Each motor has 2 wires (coin motors are not polarized — orientation doesn't matter). Wire 1 goes to the MOTOR +3.7V rail; Wire 2 goes to the corresponding ULN2803A output pin.

Motor #	Glove Location	ULN Chip	ULN Output Pin	Pin Label
0	Index tip	ULN #1	18	1C
1	Index middle	ULN #1	17	2C
2	Index base	ULN #1	16	3C
3	Middle tip	ULN #1	15	4C
4	Middle middle	ULN #1	14	5C
5	Middle base	ULN #1	13	6C
6	Ring tip	ULN #1	12	7C
7	Ring middle	ULN #1	11	8C
8	Ring base	ULN #2	18	1C
9	Pinky tip	ULN #2	17	2C
10	Pinky middle	ULN #2	16	3C
11	Pinky base	ULN #2	15	4C
12	Thumb tip	ULN #2	14	5C
13	Thumb middle	ULN #2	13	6C
14	Thumb base	ULN #2	12	7C
15	Palm upper-left	ULN #2	11	8C
16	Palm upper-mid	ULN #3	18	1C
17	Palm upper-right	ULN #3	17	2C
18	Palm lower-left	ULN #3	16	3C
19	Palm lower-right	ULN #3	15	4C

Note: glove locations are suggestions. Adjust to your preferred anatomical mapping. The electrical wiring stays the same regardless of where each motor is physically placed.

3.7 Battery connections

Wire	Connect To	Notes
18650 holder (+) red	MOTOR +3.7V rail	Through inline switch (optional but recommended)
18650 holder (–) black	Shared GND rail	MUST be same GND as ESP32

Safety: Use a protected 18650 cell from a reputable brand (Samsung, LG, Panasonic, Sony). Avoid cells marked >3500 mAh — these are almost always fake. Inline switch should break the (+) line only. Always remove the battery from the holder when not in use.

4. Assembly Notes

4.1 Recommended assembly order

Step 1: Mount IC sockets first

Solder DIP-16 (for 74LS595s) and DIP-18 (for ULN2803As) sockets onto perfboard. Match notch orientation. Do NOT insert chips yet.

Step 2: Solder power rails

Run two distinct +V rails (CHIP +5V and MOTOR +3.7V) and one shared GND rail. Use thicker wire for power lines (24 AWG or larger).

Step 3: Add decoupling capacitors

100nF ceramic between Vcc and GND of each IC, as close to the chip pins as possible. Add one 10μF electrolytic across each +V rail.

Step 4: Wire the chip-side connections

Connect ESP32 header pins, 74LS595 support pins, daisy chain wires, and 74LS595→ULN input lines. Use thin wire (28-30 AWG) for signal lines.

Step 5: Wire motor connections to ULN outputs

Solder header pins or terminal blocks for each motor pair (+ and signal). This allows easy attachment/detachment of the glove ribbon cable.

Step 6: Connect the battery system

Holder, switch, and rails. Battery NOT installed yet.

Step 7: Visual inspection

Verify all 7 connection groups from Section 3 are correctly wired. Use a multimeter to check continuity. Confirm CHIP +5V and MOTOR +3.7V rails are NOT connected.

Step 8: Insert ICs and power-on test

Insert all chips into sockets, observing notch direction. Connect ESP32 via USB. Confirm WiFi connection in serial monitor. Do NOT install battery yet.

Step 9: Test motors without battery

Verify the ESP32 responds to web commands. No motors should run (no motor power yet).

Step 10: Install battery and test motors

Insert 18650 cell, flip switch ON. Test motor 0 first. If it works, test each motor individually, then all together.

Step 11: Mount motors on glove

Only AFTER all electrical tests pass. Sew or glue motors into small fabric pouches inside the glove. Run wires through hidden channels back to the wrist board.

4.2 Critical assembly rules

CHIP +5V and MOTOR +3.7V must never be connected.

These two rails carry different voltages from different sources. A single wire bridging them will short-circuit the battery into the ESP32 and likely destroy components.

ALL grounds must be tied together.

ESP32 GND, all chip GNDs, and battery (–) must connect to one shared GND rail. Without this, the ULN2803A inputs see unstable voltages and behave randomly.

ULN2803A pin 10 (COM) goes to MOTOR +3.7V.

Connecting pin 10 to chip +5V instead is the most common mistake. It disables the internal flyback diodes and may damage the chip when motors switch off.

Match chip notch orientation.

All chips should have their notch (or pin-1 dot) facing the same direction. Inserting a chip backward into a powered socket will destroy it.

Decoupling capacitors are not optional for motors.

Motor switching creates current spikes that crash the ESP32 without proper decoupling. Place 100nF as close to each chip's power pins as possible.

Strain-relief all motor wires.

Solder joints on flexible wires fail quickly under repeated motion. Use a dab of hot glue or epoxy at each solder point on the board.

Test with multimeter before powering on.

After all soldering: check continuity from each chip's Vcc to the correct rail, from each GND pin to the GND rail, and verify there is NO continuity between the two +V rails.

5. Operation

5.1 Power-on sequence

1. Verify battery switch is OFF.
2. Plug ESP32 into USB (computer or 5V wall charger).
3. Wait for ESP32 to boot and connect to WiFi (~5 seconds). Open Serial Monitor at 115200 baud to see the assigned IP.
4. From a device on the same WiFi network, open the web GUI: [http://\[ESP32_IP\]/](http://[ESP32_IP]/)
5. Verify the GUI loads. Click any motor cell — nothing should vibrate yet (motor power is off).
6. Flip the battery switch ON. Motor power is now live.
7. Click a motor cell in the GUI — that motor should vibrate.

5.2 Power-off sequence

1. Click 'All Off' in the web GUI.
2. Flip the battery switch OFF.
3. Unplug USB from ESP32.
4. Remove the 18650 from the holder if storing for more than a day.

6. Troubleshooting

Symptom	Likely Cause	Fix
No motors respond	Battery off or dead	Check switch, charge battery
Some motors don't respond	Bad solder joint at that motor's ULN output	Reflow joint, check continuity
First 8 motors work, rest don't	Daisy chain broken at 595#1 → 595#2	Check pin 9 → pin 14 wire
First 16 work, last 4 don't	Daisy chain broken at 595#2 → 595#3	Check pin 9 → pin 14 wire
ESP32 resets when motors fire	GND not properly shared, or missing decoupling caps	Verify shared GND, add 10µF cap
WiFi keeps disconnecting	Motor noise on power lines	Add 100nF caps near every IC
All motors run continuously	ULN inputs floating, or G pin not pulled to GND	Check pin 13 to GND on all 595s
Motors weak / spinning slow	Battery low or fake/low-capacity	Recharge, replace with quality cell
Burning smell	Motor power shorting to chip power	DISCONNECT IMMEDIATELY, inspect rails

7. Quick-Reference Summary

7.1 What goes where

Connection Group	Quantity	Description
ESP32 power	2 wires	VIN → CHIP +5V; GND → GND rail
ESP32 control signals	3 wires	GPIO 23, 19, 21 to 595s
74LS595 support wiring	12 wires	4 per chip × 3 chips (Vcc, GND, SRCLR, G)
74LS595 clock/latch bus	shared	Pin 11s tied, pin 12s tied (across 3 chips)
74LS595 daisy chain	2 wires	595#1 pin 9 → 595#2 pin 14; #2 pin 9 → #3 pin 14
595 outputs → ULN inputs	20 wires	8+8+4 across the three pairs
ULN2803A support wiring	6 wires	Pin 9 → GND, pin 10 → MOTOR +3.7V (×3)
Motor wiring	40 wires	2 per motor × 20 motors
Battery wiring	2 wires	(+) → MOTOR rail via switch; (-) → GND rail
TOTAL	≈ 90+ wires	Plan ribbon cable routing carefully

7.2 Verification checklist (before powering on)

- All 3× 74LS595 chips have pin 16 → CHIP +5V
- All 3× 74LS595 chips have pin 8 → GND
- All 3× 74LS595 chips have pin 10 → CHIP +5V
- All 3× 74LS595 chips have pin 13 → GND
- All 3× 74LS595 chips have pin 11 connected to the same shared bus → ESP32 GPIO 19
- All 3× 74LS595 chips have pin 12 connected to the same shared bus → ESP32 GPIO 21
- 595#1 pin 9 → 595#2 pin 14 (data chain)
- 595#2 pin 9 → 595#3 pin 14 (data chain)
- All 3× ULN2803A chips have pin 9 → GND
- All 3× ULN2803A chips have pin 10 → MOTOR +3.7V (NOT chip +5V!)
- 20 wires connecting 595 outputs to ULN inputs (8+8+4)
- Each motor has one wire to MOTOR +3.7V rail
- Each motor has one wire to its ULN output pin
- Battery (+) → MOTOR +3.7V rail (via switch)
- Battery (-) → GND rail (same as ESP32)
- CHIP +5V rail and MOTOR +3.7V rail have NO direct connection (test with multimeter)
- All GND points are continuous (test with multimeter)
- All chips are oriented with notch in same direction
- Decoupling capacitors installed near each IC
- No exposed wires touching adjacent traces

Once all items are checked, proceed to the operation sequence in Section 5.